



AIAA 90-1568

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And Spacing Control On An Arbitrary
Number Of Boundaries**

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**AIAA 21st Fluid Dynamics, Plasma Dynamics
and Lasers Conference**

June 18-20, 1990 / Seattle, WA

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ABSTRACT

A procedure for the generation of two and quasi three dimensional grids with control of orthogonality and spacing with respect to any and/or all boundaries of the domain is described. The elliptic grid generation equations of Thompson are solved implicitly. Control of the grid behavior is achieved through the introduction of forcing function terms in the manner of Steger and Sorenson or in a modification of the method of Hilgenstock. The forcing function terms are constructed on the boundaries and propagated into the domain using transfinite Lagrangian bivariate interpolation. An anisotropic transfinite stencil is introduced and is shown to produce excellent grid behavior particularly in the vicinity of corner singularities. Emphasis is placed on the generation of viscous grids and the method is shown to be suited for use in the generation of grids for internal as well as external flow geometries. A FORTRAN program named PISCES has been written to implement the algorithm. Examples of grids for internal and external flows are given that highlight the characteristics and behavior of the algorithm.

INTRODUCTION

The generation of boundary conformal curvilinear viscous grids for internal flow geometries can routinely present difficulties that do not usually occur for external flows. These difficulties generally arise when considering a flow path having a non-rectangular cross section. Such a non-rectangular cross section can be gridded using either a polar or an H topology. The polar topology works nicely from a grid standpoint but creates a singularity in the center of the domain that requires special treatment by the flow solver. The H topology is desirable from a flow solver standpoint but causes corner singularities, as shown in figure 1, to arise. Generation of a reasonable grid in the vicinity of these corner singularities can be difficult unless propagation of the grid lines can be carefully controlled.

The corner singularity is created by the presence of the intersection of the ξ and η boundaries with an included angle approaching or exceeding 180 degrees. Unless the grid behavior local to the singularity can be controlled in a manner that takes into account the presence of both boundaries, grid cells unsuitable due to skewness and/or extremely small Jacobians, can result.

A grid generation method has been developed that provides the necessary control in the corner singularity region based on the elliptic grid generation equations of Thompson et. al. [1] and the forcing function term formulation of Steger and Sorenson [2]. In addition, a modified form of the method developed by Hilgenstock [3] has been developed that has been found to be very robust for viscous grids. These methods have been incorporated into a grid generation code PISCES which has been applied to internal as well as external flow problems.

The emphasis of this work is to present the details of the algorithm and to present typical applications that illustrate the capabilities and the characteristics of the grids that are generated.

GOVERNING EQUATIONS

The elliptic grid generation equations pioneered by Thompson et. al. [1] are employed in the current work. In vector form the equation may be written

$$\nabla^2 \xi^i = P^i \quad (i = 1, 2, 3) \quad (1)$$

Thomas and Middlecoff [4] developed a modified form of equation 1 such that the forcing function magnitudes were reduced in comparison to the form employed by others. This modified form can be expressed in two dimensional Cartesian coordinates as

$$g_{22}(\bar{r}_{\xi\xi} + P\bar{r}_{\xi}) + g_{11}(\bar{r}_{\eta\eta} + Q\bar{r}_{\eta}) - 2g_{12}\bar{r}_{\xi\eta} = 0 \quad (2)$$

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where:

$$\begin{aligned}
 \vec{r} &= x\hat{i} + y\hat{j} \\
 g_{11} &= x_\xi^2 + y_\xi^2 \\
 g_{22} &= x_\eta^2 + y_\eta^2 \\
 g_{12} &= x_\xi x_\eta + y_\xi y_\eta \\
 P &= \text{forcing function acting in } \xi \text{ direction} \\
 Q &= \text{forcing function acting in } \eta \text{ direction}
 \end{aligned} \tag{3}$$

and the Greek ξ and η subscripts indicate a partial derivative of the variable with respect to ξ or η .

Equation 2 represents a system of coupled nonlinear partial differential equations (PDEs). This equation set is elliptic and requires the solution of a boundary value problem. A popular approach to solve elliptic equations is to employ an implicit relaxation procedure. Common approaches to date have been symmetric successive over-relaxation (SSOR) and successive line over-relaxation (SLOR). In addition, multi-grid procedures have been employed by several authors [5,6]. The current work has employed a semi-implicit method developed by Peric [7]. This method uses partial cancellation in the manner of Stone [8] to develop an iterative approximate LU factorization of the coefficient matrix. This procedure was chosen because it allows the linearized and discretized form of equation 2 to be solved such that the right hand side of the linear algebraic equations are zero. The procedure has been found to be extremely robust and reliable.

FORCING FUNCTION METHODOLOGY

Three forcing function methods have been investigated and compared to determine the most suitable approach for a wide class of problems. These are the method of Thomas and Middlecoff [4], Sorenson and Steger [2], and Hilgenstock [3]. The first method is a non-iterative forcing function approach in the sense that the functions are computed once and held fixed throughout the course of obtaining a solution to the nonlinear PDEs. The second and third method are iterative procedures that adjust the boundary forcing functions to meet a prescribed angle and spacing constraint at the boundary. All three procedures require the interpolation of the forcing function into the domain based on the boundary values. The interpolation procedures will be discussed in a separate section to follow.

The forcing function method derived by Thomas and Middlecoff was obtained from equation 2 for a given coordinate boundary by "imposing two a priori constraints on the local slope and curvature of the coordinate curves transverse to that boundary" [4]. This approach thereby prescribes one of the forcing functions per boundary, i.e.

P is computed on η constant boundaries and Q is computed on ξ constant boundaries. The forcing functions from [4] are computed from

$$\begin{aligned}
 P &= -\frac{x_\xi x_{\xi\xi} + y_\xi y_{\xi\xi}}{g_{11}} \\
 Q &= -\frac{x_\eta x_{\eta\eta} + y_\eta y_{\eta\eta}}{g_{22}}
 \end{aligned} \tag{4}$$

The method developed by Sorenson and Steger requires that both P and Q be computed on any given boundary for control of the grid angle and spacing of the first grid point away from the wall. The derivation can be found in [3] and will not be presented here. The equations that determine P and Q have the form

$$\begin{aligned}
 P &= -\frac{\vec{r}_\xi \cdot \vec{r}_{\xi\xi}}{g_{11}} - \frac{\vec{r}_\xi \cdot \vec{r}_{\eta\eta}}{g_{22}} \\
 Q &= -\frac{\vec{r}_\eta \cdot \vec{r}_{\eta\eta}}{g_{22}} - \frac{\vec{r}_\eta \cdot \vec{r}_{\xi\xi}}{g_{11}}
 \end{aligned} \tag{5}$$

The method of Hilgenstock has a very different form compared to that of Sorenson and Steger's but is similar in the grid control obtained. Hilgenstock extended the concept of Von Lavante [9] to include both grid angle and spacing control. This method exploits the influence of the forcing functions on the grid in the following manner. Figure 2 illustrates that P controls the grid angle Θ relative to a constant η boundary and grid spacing ds relative to a constant ξ boundary. Conversely Q controls the grid angle relative to a constant ξ boundary and grid spacing relative to a constant η boundary. The forcing functions are then incremented from some initial value based on the local relative angle and spacing error. The update procedure then becomes

$$\begin{aligned}
 P^{n+1} &= P^n + \Delta P^{n+1} \\
 Q^{n+1} &= Q^n + \Delta Q^{n+1}
 \end{aligned} \tag{6}$$

where the ΔP and ΔQ are computed depending on whether they are controlling the angle or spacing at a given boundary. The ΔP and ΔQ for an η_{min} boundary are computed in the following manner.

$$\begin{aligned}
 \Delta P^{n+1} &= -\tan^{-1}\left(\frac{\theta_{req} - \theta_{actual}}{\theta_{req}}\right) \\
 \Delta Q^{n+1} &= \tan^{-1}\left(\frac{ds_{req} - ds_{actual}}{ds_{req}}\right)
 \end{aligned} \tag{7}$$

The use of the $\tan^{-1}$ function is an artifice that controls the maximum value of the delta while smoothly decreasing the allowable delta as the error is reduced. A modified form of this approach has been developed that has been found to increase the convergence rate as much as 40 percent. The modification results in equations 7 being

rewritten as

$$\begin{aligned}\Delta P^{n+1} &= -\tanh\left(\frac{\theta_{req} - \theta_{actual}}{\theta_{req}}\right)\beta \\ \Delta Q^{n+1} &= \tanh\left(\frac{ds_{req} - ds_{actual}}{ds_{req}}\right)\beta\end{aligned}\quad (8)$$

A hyperbolic function is used to ensure that the Δ will always approach unity asymptotically whereas the β allows the slope of the function at the origin to be controlled to increase the magnitude of Δ for small relative errors. A value for β of approximately .8 has been found to yield substantial improvements with little effect on robustness. Figure 3 shows the change in the Δ function from a β of 1.0 to .8.

INTERPOLATION METHODOLOGY

Interpolation of the forcing functions from the boundaries can play a very important role in the quality of the grid obtained and in the stability and robustness of the solution procedure. Other works have used one dimensional interpolation functions [2] or averaged multiple one dimensional interpolation functions [3]. However, when considering grid topologies typical of internal flows multiple viscous boundaries are the rule. When these boundaries intersect a careful blending of the forcing functions must be devised to create reasonable grid behavior and a robust solution procedure. The approach used in the current work is to employ transfinite interpolation as the basis of the blending of the forcing functions from all four boundaries. A derivation of transfinite interpolation can be found in [10] and will not be repeated here. The general two-dimensional form may be written as

$$F = F_\xi + F_\eta - F_{\xi\eta} \quad (9)$$

where:

$$\begin{aligned}F_\xi &= \text{unidirectional interpolation function in } \xi \\ F_\eta &= \text{unidirectional interpolation function in } \eta \\ F_{\xi\eta} &= \text{correction interpolation function}\end{aligned}\quad (10)$$

These interpolation functions can be written for any of several types of interpolation. Lagrangian and Hermitian polynomials, or even splines are possible candidates for use. For the sake of simplicity only Lagrangian functions of degree 3 or less have been considered. A general expression for a Lagrangian polynomial of arbitrary degree has been given in Thompson et. al. [11] as

$$F_\xi = \sum_{n=1}^N \Phi_n\left(\frac{\xi}{I}\right) F_n \quad (11)$$

where:

$$\Phi_n\left(\frac{\xi}{I}\right) = \prod_{l=1}^N \frac{\xi - \xi_l}{\xi_n - \xi_l} \quad (l \neq n) \quad (12)$$

Equation 9 can then be rewritten for a Lagrangian polynomial as

$$\begin{aligned}F &= \sum_{n=1}^N \Phi_n\left(\frac{\xi}{I}\right) F(\xi_n, \eta) + \sum_{m=1}^M \Psi_m\left(\frac{\eta}{J}\right) F(\xi, \eta_m) \\ &\quad - \sum_{n=1}^N \sum_{m=1}^M \Phi_n\left(\frac{\xi}{I}\right) F(\xi_n, \eta_m) \Psi_m\left(\frac{\eta}{J}\right) F(\xi_n, \eta_m)\end{aligned}\quad (13)$$

For a Lagrangian polynomial of degree 2 ($N=2$ and $M=2$), the blending functions $\Phi(\frac{\xi}{I})$ and $\Psi(\frac{\eta}{J})$ are

$$\begin{aligned}\bar{\xi} &= \left(\frac{\xi - \xi_{min}}{\xi_{max} - \xi_{min}} \right) \\ \Phi_n\left(\frac{\xi}{I}\right) &= \begin{cases} 1 - \bar{\xi} & n = 1 \\ \bar{\xi} & n = 2 \end{cases}\end{aligned}\quad (14)$$

$$\begin{aligned}\eta &= \left(\frac{\eta - \eta_{min}}{\eta_{max} - \eta_{min}} \right) \\ \Psi_m\left(\frac{\eta}{J}\right) &= \begin{cases} 1 - \bar{\eta} & m = 1 \\ \bar{\eta} & m = 2 \end{cases}\end{aligned}$$

For a Lagrangian polynomial of degree 3 ($N=3$ and $M=3$), the blending functions $\Phi(\frac{\xi}{I})$ and $\Psi(\frac{\eta}{J})$ are

$$\begin{aligned}\bar{\xi} &= \left(\frac{\xi - \xi_{min}}{\xi_{max} - \xi_{min}} \right) \\ \Phi_n\left(\frac{\xi}{I}\right) &= \begin{cases} 2 \left[\bar{\xi} - \frac{1}{2} \right] [\bar{\xi} - 1] & n = 1 \\ 4 [\bar{\xi}] [1 - \bar{\xi}] & n = 2 \\ 2 [\bar{\xi}] \left[\bar{\xi} - \frac{1}{2} \right] & n = 3 \end{cases}\end{aligned}\quad (15)$$

$$\begin{aligned}\eta &= \left(\frac{\eta - \eta_{min}}{\eta_{max} - \eta_{min}} \right) \\ \Psi_m\left(\frac{\eta}{J}\right) &= \begin{cases} 2 \left[\bar{\eta} - \frac{1}{2} \right] [\bar{\eta} - 1] & m = 1 \\ 4 [\bar{\eta}] [1 - \bar{\eta}] & m = 2 \\ 2 [\bar{\eta}] \left[\bar{\eta} - \frac{1}{2} \right] & m = 3 \end{cases}\end{aligned}$$

The polynomial of degree 3 is extremely useful for grid generation because it allows the effect of the boundaries

to be decayed to zero in the center of the domain. This is accomplished by setting the n and/or m equal 2 coefficients to zero. Doing so causes the forcing function being interpolated to approach zero in the center of the domain making the PDEs locally Laplacian. This decouples the boundaries and results in improved stability and robustness of the overall algorithm.

The definition of transfinite interpolation does not preclude the possibility of using a different polynomial in each of the coordinate directions. This feature has been exploited in the following manner. When interpolating a forcing function into the domain a degree 2 polynomial is used in the direction normal to the direction of action of the forcing function. A polynomial of degree 3 is used in direction tangential to the direction of action of the forcing function. This causes the angle constraint to propagate linearly from boundary to boundary and the spacing constraint to decay to zero at center of the domain. Figure 4 illustrates this anisotropic transfinite interpolation procedure as compared with an isotropic procedure. The motivation for this approach was the observation that isotropic interpolation using a polynomial of degree 2 could cause the algorithm to enter a limit cycle. This difficulty was determined to be due to the forcing functions controlling the spacing of the grid. Experimentation showed that increasing the decay rate of the function would improve convergence and stability. If required the polynomial of degree 3 could be replaced with any higher order function to modify the decay behavior.

Having addressed the unidirectional polynomials the remaining coefficients in equation 13 due to the correction interpolation function need to be determined. Determination of these coefficients requires the forcing functions be known in the corner region. However, equations 5 or 6 and 7 do not provide for an unambiguous procedure in the corners. The ambiguity can be alleviated by using a linear weighted average of the values adjacent to the corner. Linear averaging has proven sufficiently general for all grids computed to date.

RESULTS

The methods described above have been applied to internal and external flow configurations with equal success. The anisotropic blending function approach to transfinite interpolation has been found to be equally suitable for internal external flow grids and overall has proven to be the most reliable.

An ellipse with an aspect ratio of 2 and a constant minimum wall spacing of .001 has been used to serve as a shape typical of internal flow problems. The grids generated were 75 x 50 for a total of 3750 grid points. Three forcing function methods have been examined; Thomas and Mid-

dlecoff, Sorenson and Steger, and modified Hilgenstock.

Figure 5 presents the grid produced using the Thomas and Middlecoff forcing functions. The grid behavior near the boundaries is non-orthogonal and the singularity is highly skewed. The method did, however, converge in 25 iterations and has been found to be very stable and robust. Figure 6 presents the same grid generated using isotropic degree 3 Lagrangian transfinite interpolation and the Sorenson and Steger forcing functions. The orthogonality in the near wall region has improved significantly and the skewness near the singularity is reduced. The influence of the degree 3 polynomial is most visible in the manner in which the orthogonality decays as the grid approaches the center of the domain. This decay is a result of the polynomial causing the forcing functions to decay to zero at the center of the domain. The orthogonality can be propagated further into the domain by employing either a isotropic degree 2 or an anisotropic degree 2/3 transfinite interpolation of the forcing functions. The isotropic degree 2 case has been found to result in poor convergence. However, the anisotropic method has been found to converge efficiently and results in the desired grid characteristics. Figure 7 demonstrates the increased propagation of the orthogonality caused by the anisotropic interpolation. The grids in figures 6 and 7 required 200 iterations to reduce the RMS change in the normalized Jacobian three orders of magnitude.

The modified Hilgenstock method was also employed for the ellipse test case. Figure 8 shows the grid which was obtained using the anisotropic transfinite interpolation method. The grid behavior is very similar to that of the grid in figure 7. However, a slight degradation of orthogonality near the singularity was observed. The degradation is not surprising since the method does not contain any way for the grid curvature near the boundary to affect the forcing functions constructed at the boundary. This deficiency has not been found to cause problems for any application to date. The convergence for this case was 3.8 orders of magnitude in 200 iterations. The method in general has been found to be significantly better behaved than when using Sorenson and Steger forcing functions.

The two dimensional algorithm described above has been applied to quasi- three-dimensional grids by incorporating a procedure that maps an arbitrary $X Y Z$ plane into an local $x' y'$ plane that is solved two dimensionally and then mapped back. A three dimensional view of the grid for a rectangular to round transition duct is shown in figure 9. Five planes typical of the grid obtained are shown. The anisotropic transfinite interpolation function was used for this case due to clustering and orthogonality being required on adjacent boundaries. An end view of an η constant and a ζ constant grid plane is shown in fig-

ure 10. The grid can be seen to be very uniform in the ξ direction even though the grid was generated as discrete ξ planes.

Grids for external flows have also been generated using the algorithm. A turbulent flow grid around an RAE 2822 airfoil is shown in figures 11a - 11d. The grid was 385 x 97 with a minimum wall spacing of 1×10^{-7} and maximum aspect ratio of 1×10^5 . The grid was computed using the anisotropic interpolation of modified Hilgenstock forcing functions. The grid required 125 iterations to reduce the residual 3 orders of magnitude. At this level of convergence the mean orthogonality and spacing errors on all the boundaries were less than .1 percent. The magnified views show good behavior of the grid near the leading and trailing edges and near wake region.

Figures 12 and 13 show views of a grid for an Euler analysis of the NASA Langley High Speed Civil Transport concept vehicle [12]. The vehicle body is shown as a shaded surface with four grid cross sections superimposed. The surface geometry and discretization were obtained from a CAD geometrical modeling code and the initial grid was generated by an algebraic homotopy procedure [13] external to PISCES. The grid shown was generated using the anisotropic transfinite interpolation function and modified Hilgenstock forcing functions.

The code has also been extended to handle multi-block configurations allowing no continuity, $C(0)$ or $C(1)$ continuity across block interfaces depending on the forcing functions used on the boundaries. In addition, the planar mapping procedure has been extended to include non planar sections that have unique planar projections. Figure 14 illustrates both these features in a close up view of a 4 block grid for an actuated control strake on the nose of an F-18. This grid was developed from an initial algebraic grid from reference [14].

CONCLUSIONS

A grid generation algorithm based on the Poisson equation which allows grid orthogonality and spacing control with respect to any or all boundaries has been developed. A modification of the forcing function method of Hilgenstock was introduced that improves convergence rates for some problems on the order of 40 percent. The development of isotropic and anisotropic transfinite interpolation methodologies for the interpolation of the forcing functions was presented. This approach was shown to produce good grid behavior even in the presence of corner singularities. Examples of two- and quasi-three-dimensional internal flow grids were presented. In addition the algorithm was shown to have retained sufficient generality to be applied with equal success to external flow configurations. Although not discussed herein the transfinite in-

terpolation procedure is extendible to three dimensions. This work is currently underway.

ACKNOWLEDGEMENTS

Algorithm development was initiated while the author was a researcher at Pratt & Whitney in West Palm Beach, Florida. Development has continued and has been supported by the Theoretical Flow Physics Branch of the Fluid Mechanics Division of NASA Langley Research center under contract NAS1-18599. The author would like to express his gratitude to those persons who have continued to provide advice and assistance throughout the course of this research.

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FIGURES

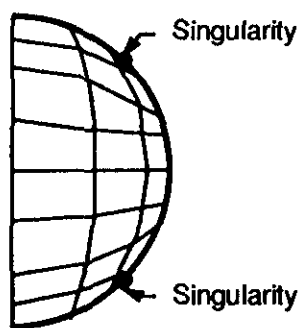


Figure 1: Corner singularity due to H topology.

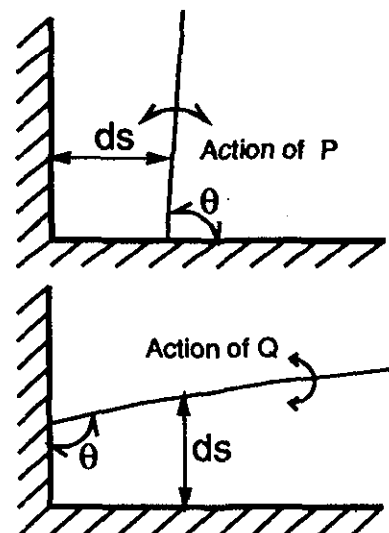


Figure 2: Direction of action of forcing fuctions.

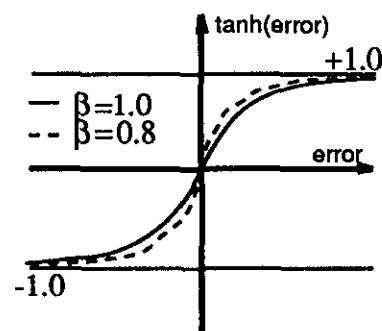


Figure 3: Behavior of $\tanh$ with respect to β .

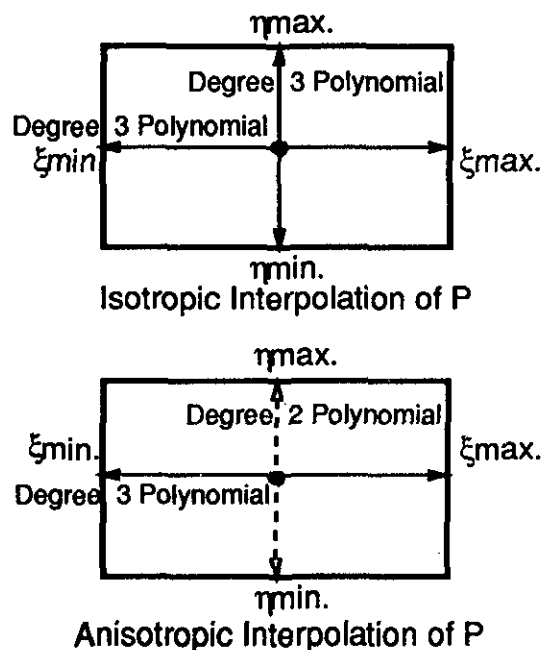


Figure 4: Illustration of isotropic and anisotropic interpolation.

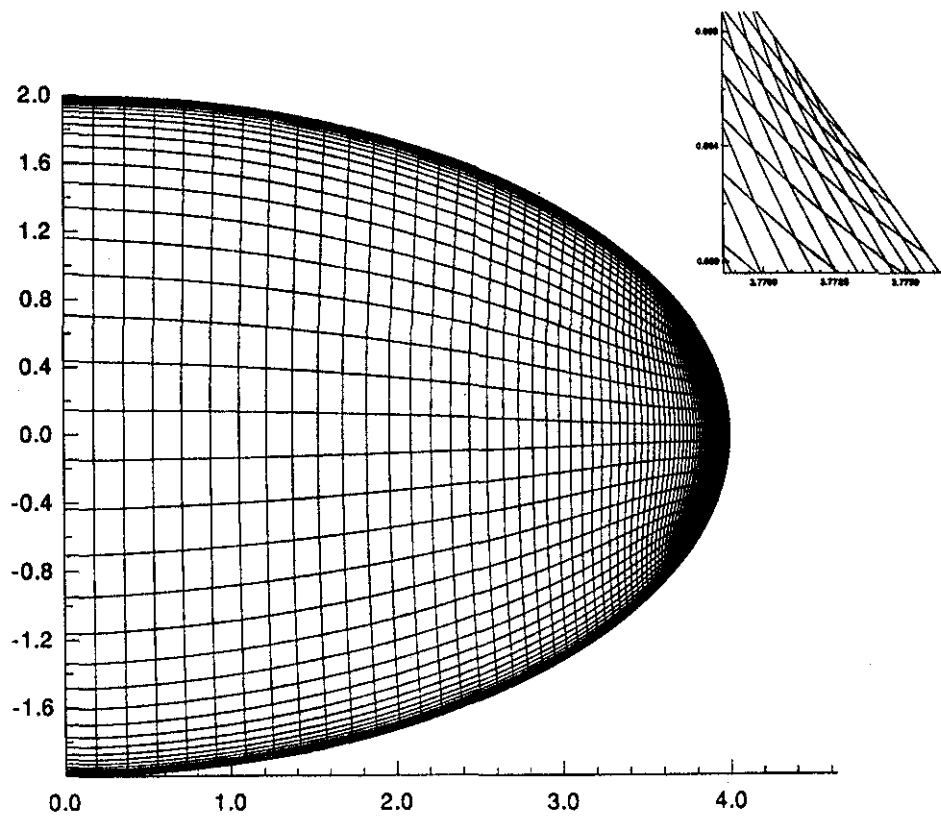


Figure 5: Fine grid in an ellipse using Thomas and Middlecoff forcing functions.

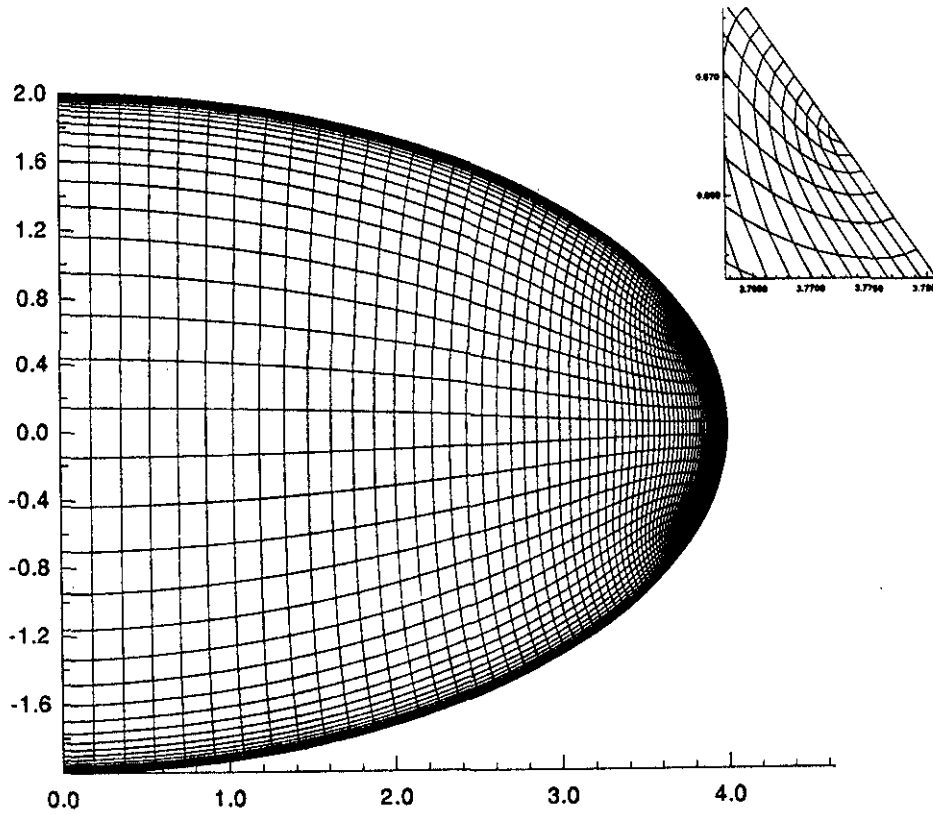


Figure 6: Fine grid in an ellipse using isotropic degree 3 interpolation and Sorenson and Steger Forcing functions.

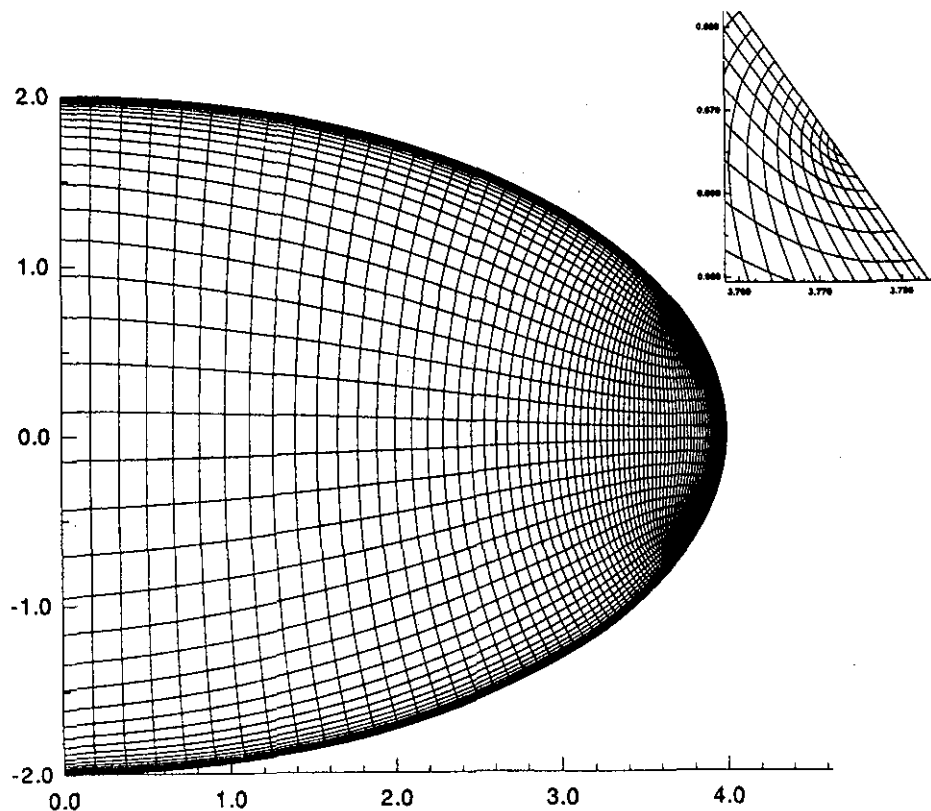


Figure 7: Fine grid in an ellipse using anisotropic interpolant and Sorenson and Steger Forcing functions.

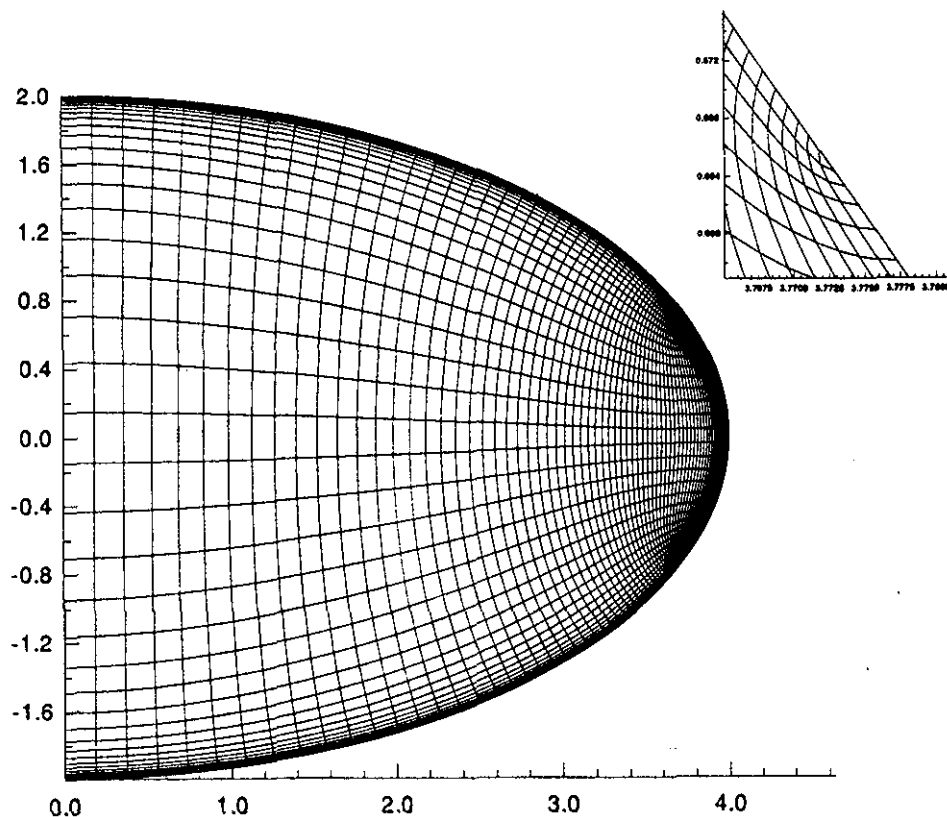


Figure 8: Fine grid in an ellipse using anisotropic interpolant and modified Hilgenstock Forcing functions.

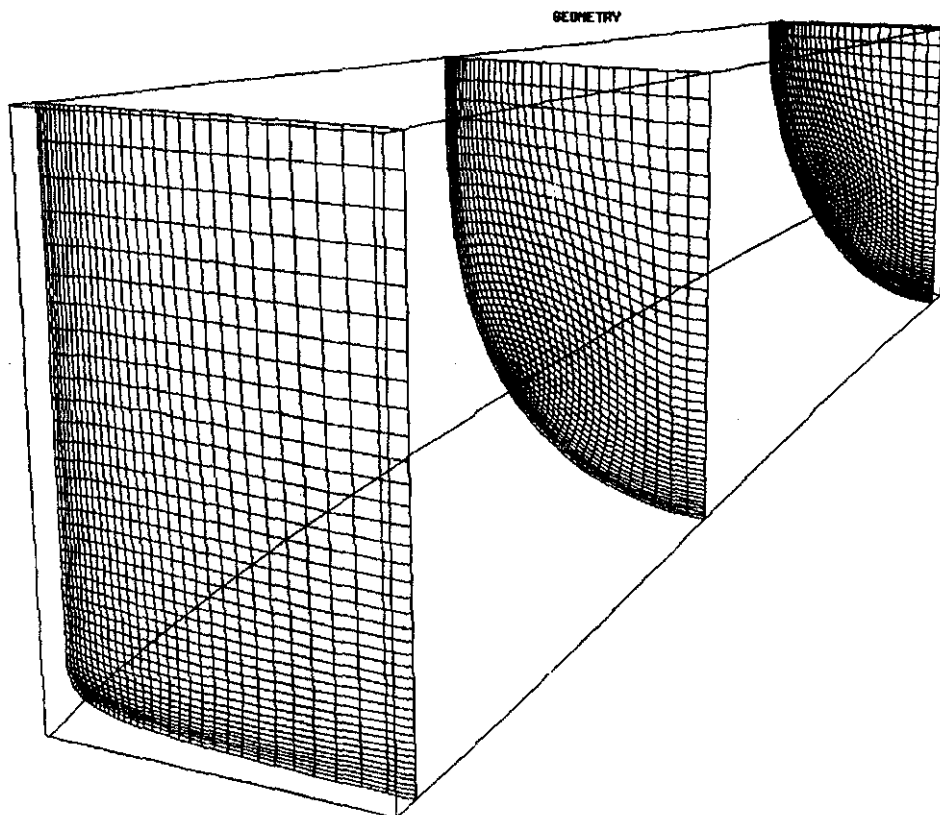


Figure 9: Rectangular to round transition duct with typical ξ grid planes.

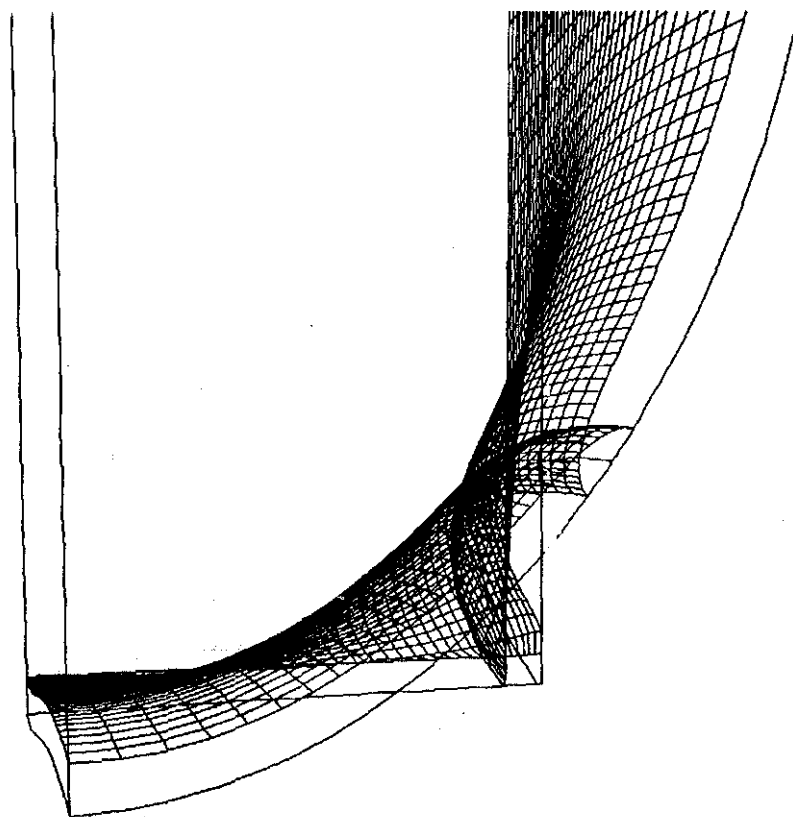


Figure 10: Rectangular to round transition duct η and ζ grid planes.

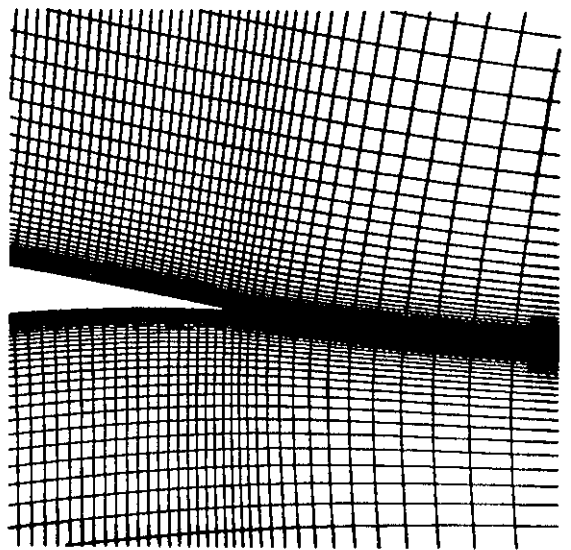
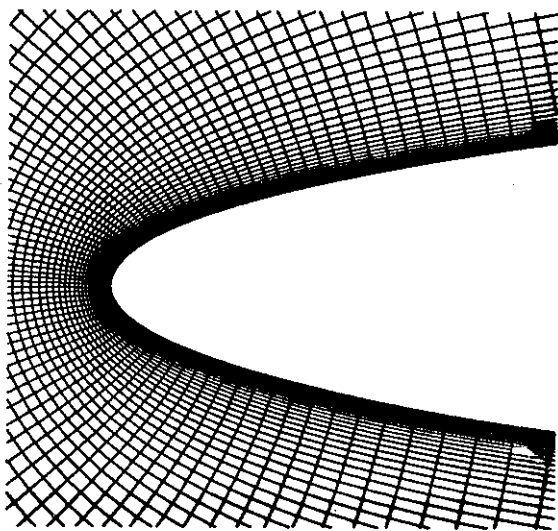
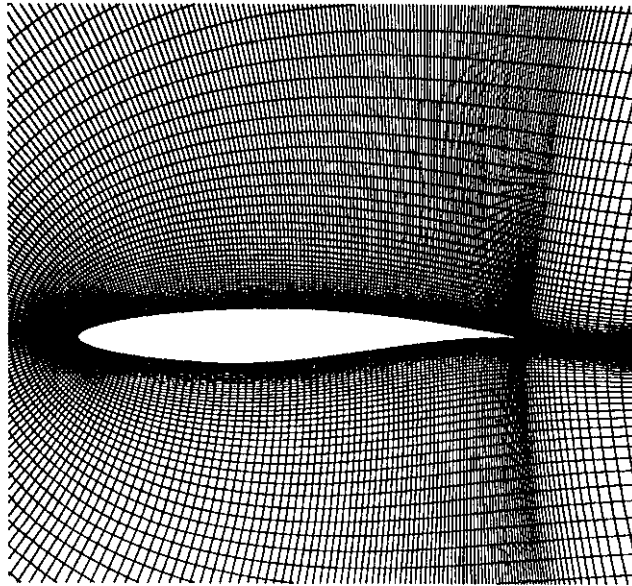


Figure 11: Viscous grid for RAE2822 AIRFOIL .

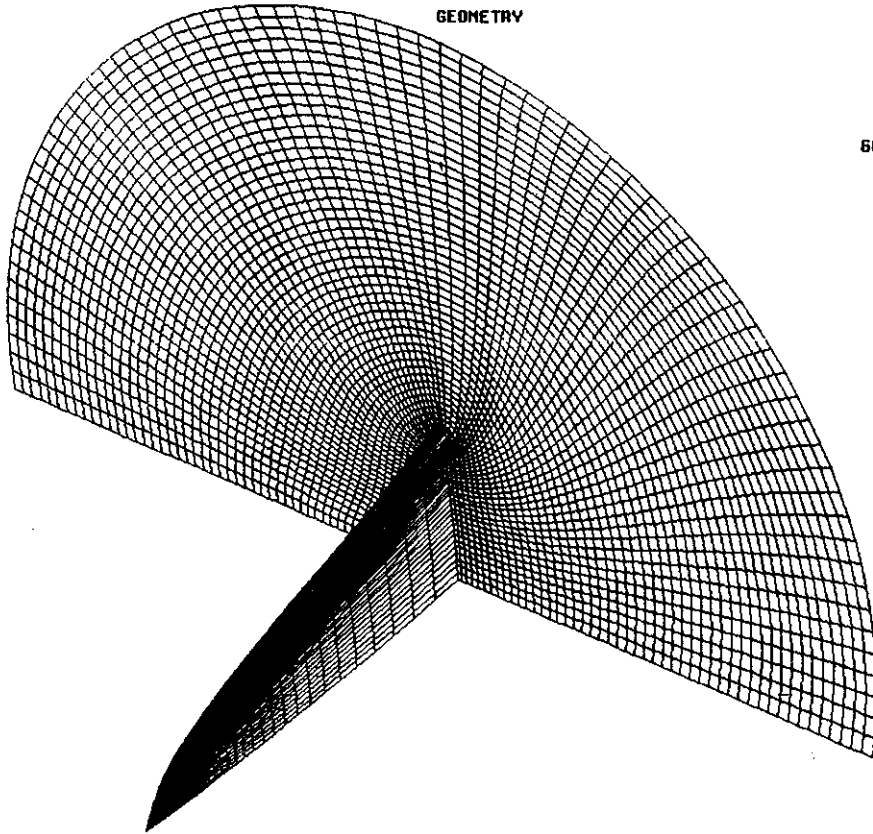


Figure 12: High Speed Civil Transport vehicle with typical ξ grid plane.

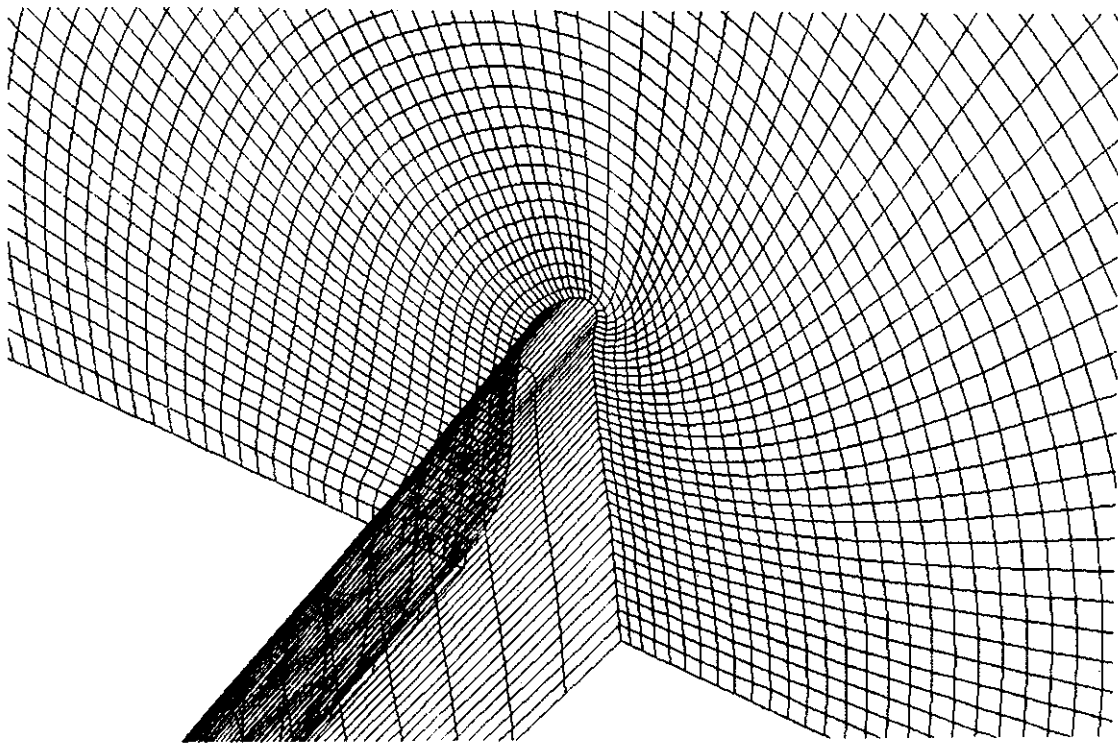


Figure 13: High Speed Civil Transport vehicle magnified view.

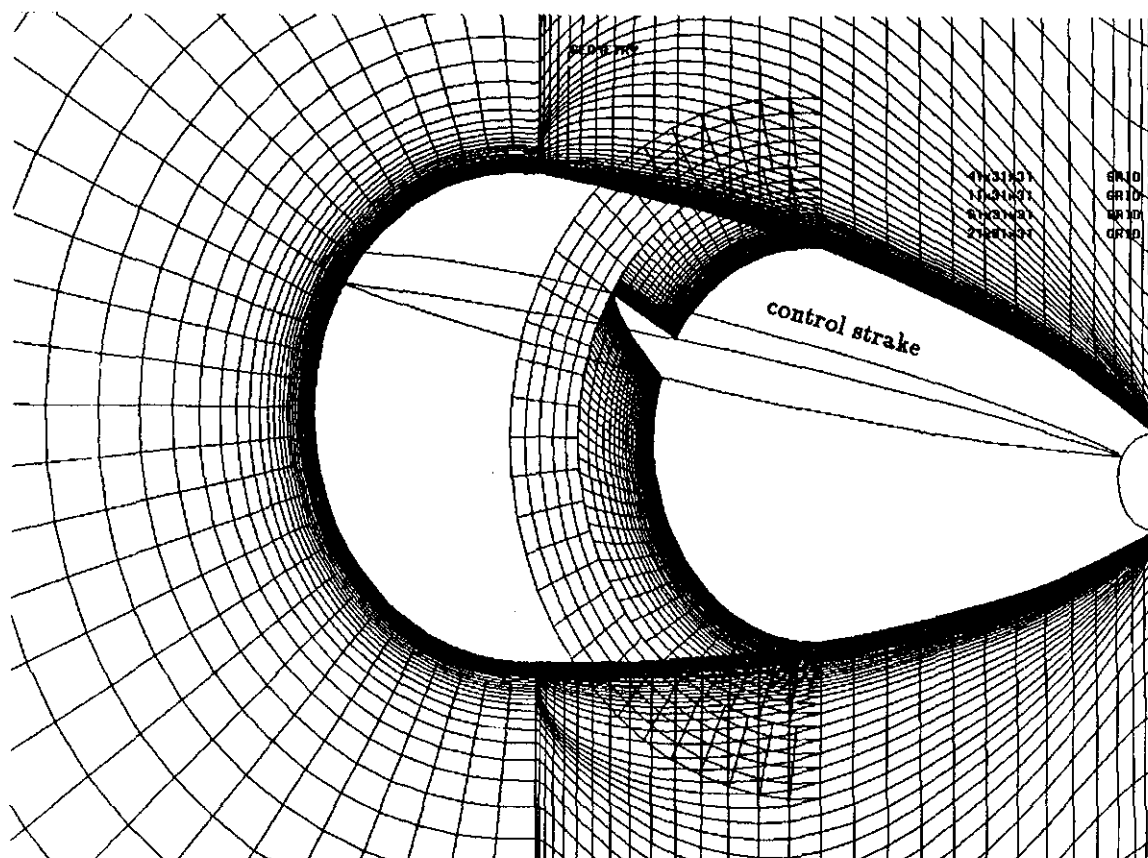
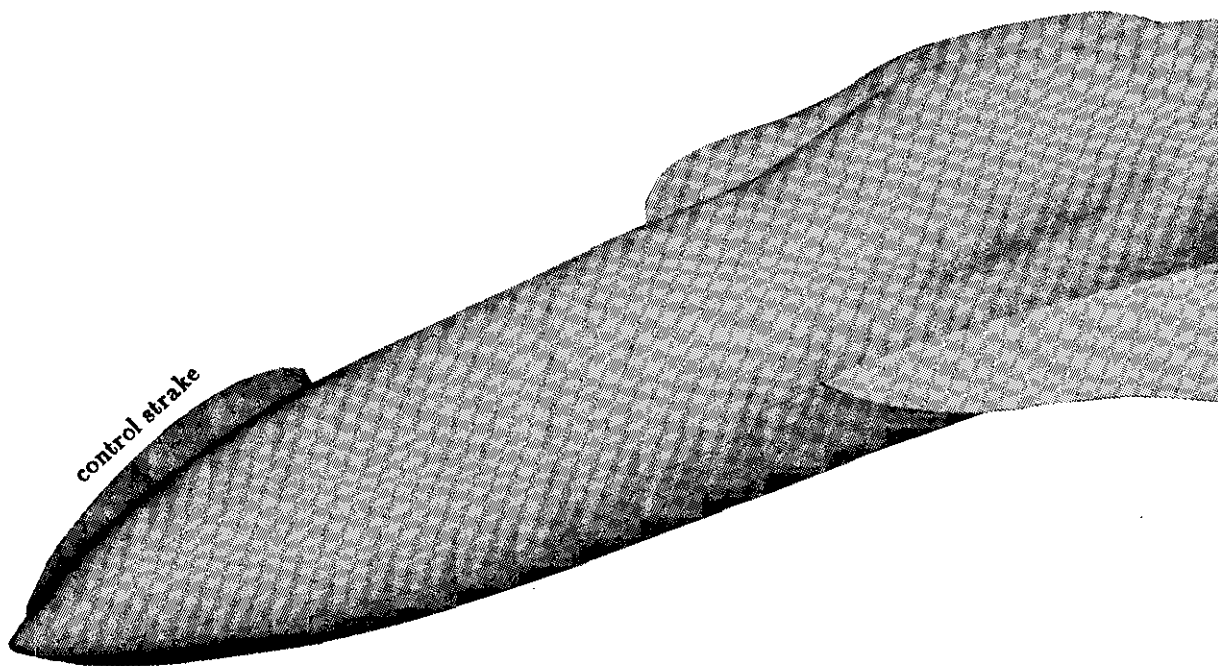


Figure 14: F18 actuated control strake multiblock grid.